

INTEGRATED INTELLIGENT SYSTEMS · LECTURE 1

Robots That Do the Task

How can we explain and design robots that accomplish everyday tasks? This course works toward a science and engineering of intelligent robot action.

Robot agency is a relationship between programs, problems, bodies and worlds.

MICHAEL BEETZ · AICOR BOOK, PREFACE AND CHAPTER 1

AICOR · INSTITUTE FOR ARTIFICIAL INTELLIGENCE · UNIVERSITY OF BREMEN

ILLUSTRATIVE TASK SCENARIO · INDIVIDUAL PROMPT

Eva's coffee request leaves action decisions open

Eva asks, “Bring me my coffee.” Name the goal, two constraints and two unknowns.



Her usual blue mug is in the dishwasher. The white mug is available. What must the robot ask or observe before it commits?

Goal	Constraints	Unknowns

AI-generated scene with schematic annotations · illustrative home-assistance case

WARM-UP · WRITTEN ANSWERS · REVISITED AT THE END OF THE LECTURE

Warm-up questions

Q1 Which can robots demonstrate today: repeat a factory task, learn to handle several household objects, or assist someone autonomously in an unfamiliar home?

Q2 When you ask a generative model a question, what does it give you back?

Q3 A video shows a robot succeeding at a task 100 times in a row. Can that robot do the task?

LECTURE 1 · OUTCOMES

Learning outcomes

1. **Explain** why everyday manipulation is the test case for intelligence, and the standard the Nilsson challenge sets for a robot.
2. **Describe** how today's AI-powered robots work and what data alone supplies; **explain** what it means to understand a robot site in terms of five kinds of question.
3. **State** the body motion problem for an everyday request: what must be achieved, and what prevented.
4. **Explain** the relationship between control programs, manipulation problems, embodiments and worlds, and how models make claims about it testable.

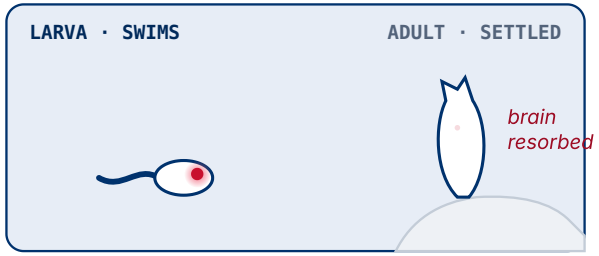
This course assumes Python, conditional probability, reading a first-order logic formula, and coordinate frames — none needs to be strong today; each becomes necessary in a later lecture (Python from Sheet 1; probability from Lecture 3; frames in Lectures 4–5 and 12; logic in Lecture 13).

THE GRAND RESEARCH CHALLENGE

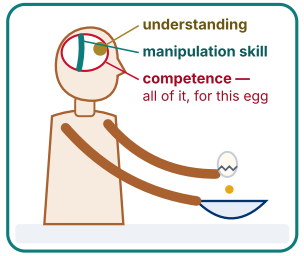
Brains move bodies to change the world and satisfy goals



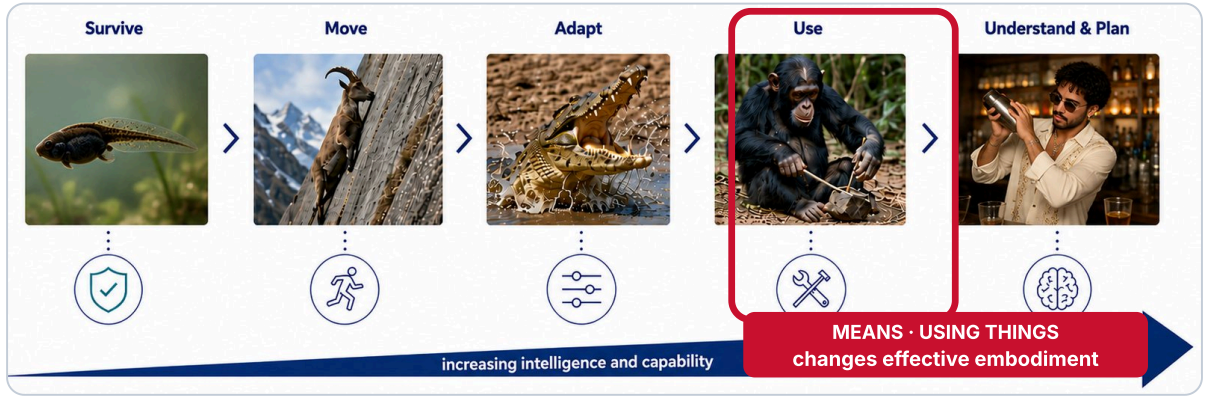
never moves
no brain



swims and steers with a brain, settles, then loses it
the sea squirt



cracking an egg into a bowl
a brain built for using things



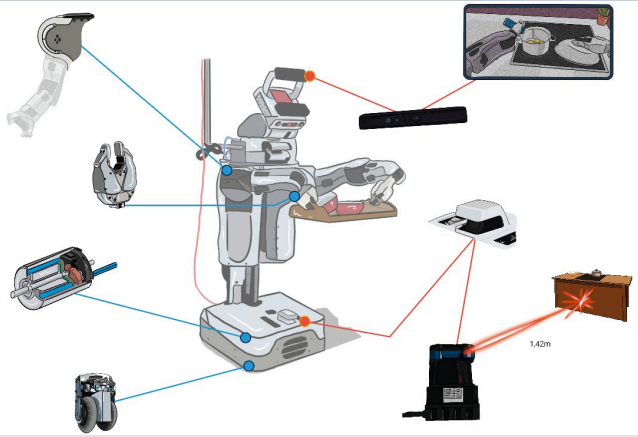
THE GRAND CHALLENGE

Achieve changing goals **flexibly and reliably in an open environment**: determine what success means in each context, transfer to novel goals and situations, and use things to change effective embodiment.

AI-generated illustrations — tree, capability ladder · person cracking an egg and animation: drawn for this lecture; brain regions simplified

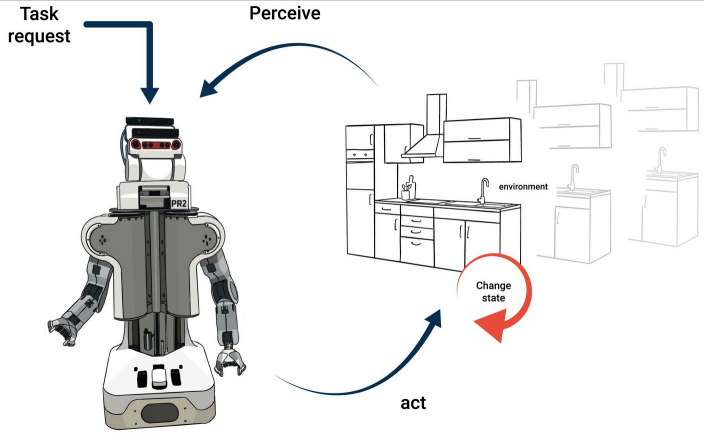
THE GRAND CHALLENGE, FOR MACHINES

A robot is an articulated device whose body is moved by a program



The body — links joined by motor-driven joints; sensors for the body and the world

illustration · intel4coro textbook · IAI Bremen



The loop — a task request comes in; the robot perceives the kitchen, acts, and the world changes

illustration · intel4coro textbook · IAI Bremen

DEFINITION · ROBOT

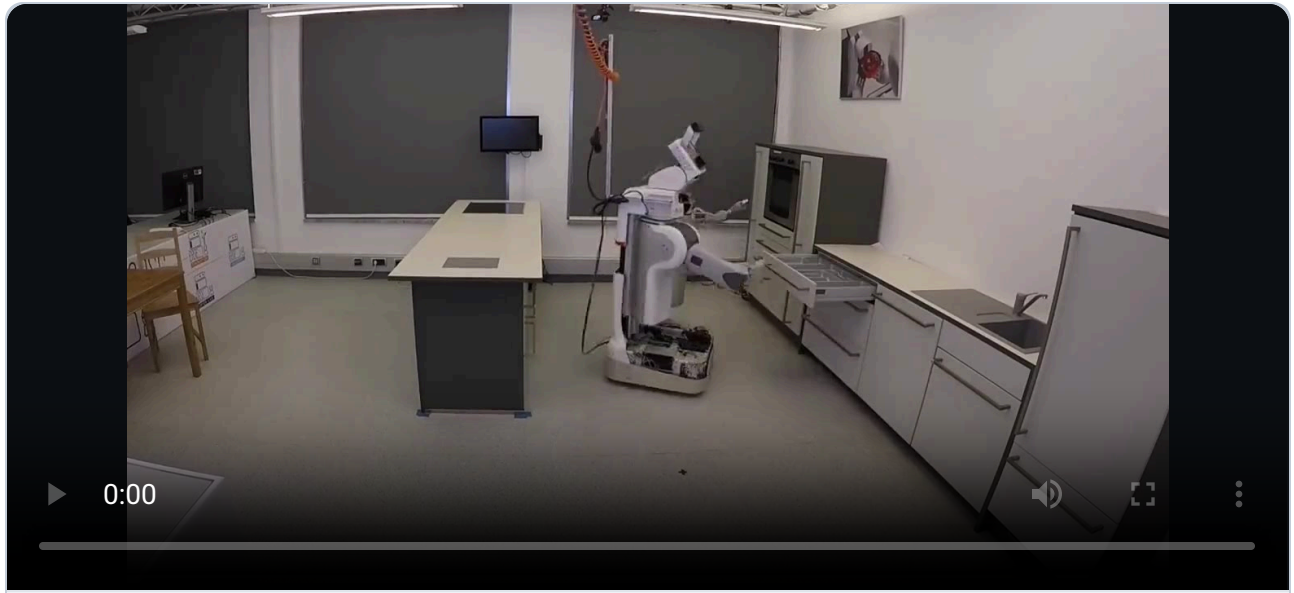
A **robot** is an articulated electro-mechanical device that is operated and controlled by computer programs in order to accomplish tasks.

WHERE THE DIFFICULTY LIES

The program closes the loop: it reads the request and what the sensors report, and chooses motions that bring about what was asked and avoid what must not happen.

THE NILSSON CHALLENGE

The challenge: a robot should master tasks its body and sensors can support



Setting the table — PR2 in the IAI apartment lab
IAI · University of Bremen

First build — teleoperation illustration · **second build** — table-setting video

AFTER NILSSON'S EMPLOYMENT TEST (2005)

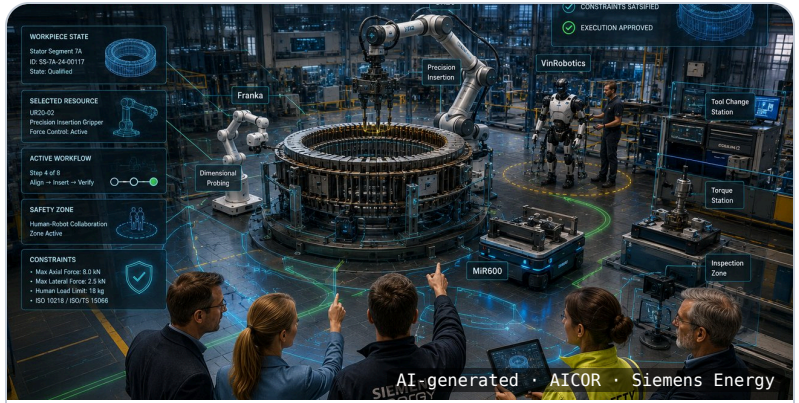
A robot is to be programmed to do what we can reasonably expect of it, given its sensors, its body and its effectors: roughly the tasks a person could accomplish by controlling the robot with a game controller.

RATIONALE FOR THIS STANDARD

Teleoperation shows what this body can do under the operator's conditions. Autonomy also needs sensing, task interpretation and control that the robot can obtain and use itself.

WHERE WE WANT TO GO · THREE VISIONS

Three application visions concern production, the research laboratory and the home



PRODUCTION · WIND-TURBINE GENERATORS

“Insert stator segment S-10235 into slot 14 — seat it, secure it, verify it.”

- every segment measured: +0.08 mm changes the forces
- at most 340 N; insulation surfaces untouched
- a quality record for every step



RESEARCH LAB · MICROBIAL CULTURE PLATES

“Read every plate overnight, in its window — with nobody in the room.”

- every plate, every incubation, every SOP revision
- sterility unbroken; at most 5 N; read within 18–22 min
- an auditable trail for every reading



HOME · EVA’S MORNING, GARMI CHALLENGE

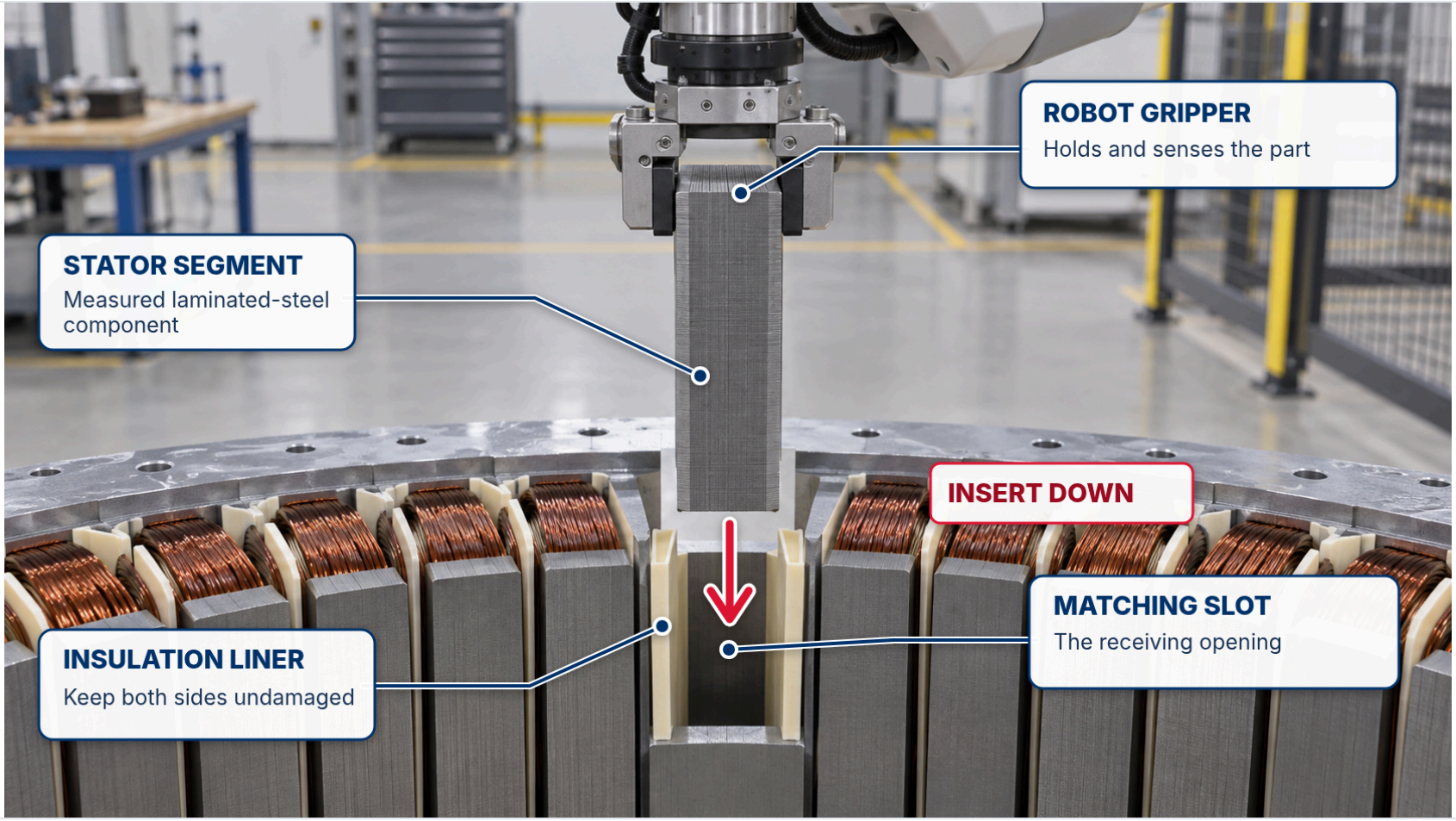
“Bring me coffee, then make this place a little less chaotic.”

- every home, every morning, every request
- no scalding; walking frame free; medication untouched
- a record of what it moved, and why

no two situations alike · **hard limits that must hold** · **evidence that it was right**

PRODUCTION VISION · THE PHYSICAL TASK

Seat the measured stator segment in its matching slot



Align → lower → monitor contact. This scenario limits force to 340 N and requires an episode record.

AI-generated illustrative scene with schematic annotations · geometry and limits belong to the lecture vignette

AICO | LECTURE 1 · FIVE ACTS

Overview

1 **AI robots today**
What robots demonstrate, how they learn action, and whether data suffices.

2 **What understanding means**
Which programs solve which problems, for which bodies and worlds — and how we know.

3 **The smallest case**
Return to Eva's request through one fragile egg grasp.

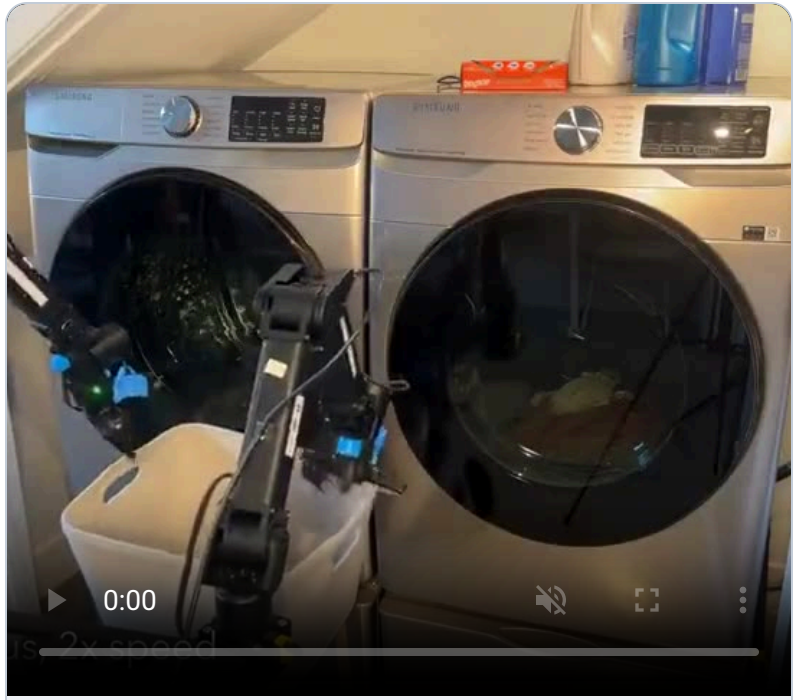
4 **Designing the solution**
The integration diagram and the warrant gate, where proposals become commitments.

5 **The course**
The map of the semester and the work of our lab.

ACT 1 · AI ROBOTS TODAY

Robots unload laundry, dance with swords, and drive

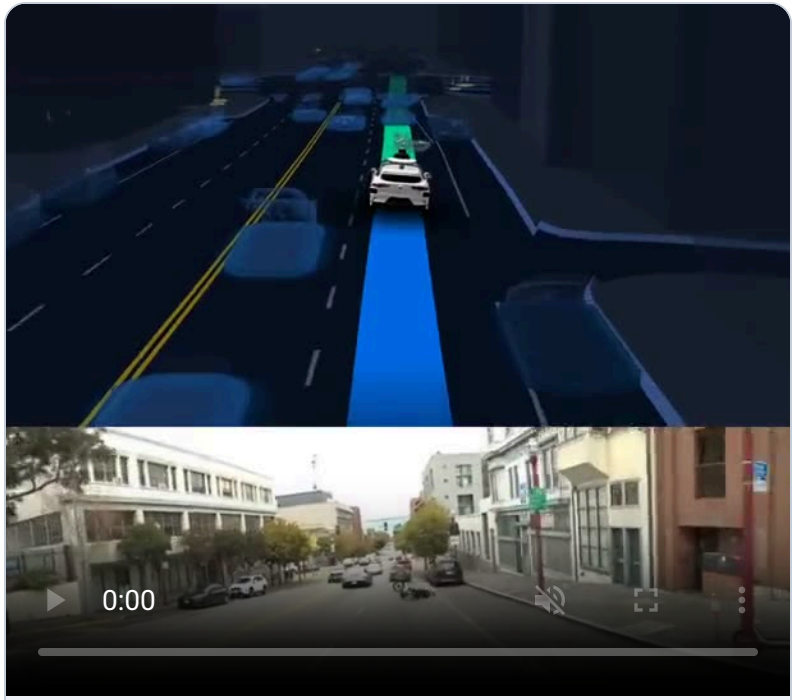
Three impressive demonstrations — each invites a question about the conditions under which it works.



Unloading a dryer
Physical Intelligence (π_0)



Sword dancing
CCTV Spring Festival Gala · euronews No Comment



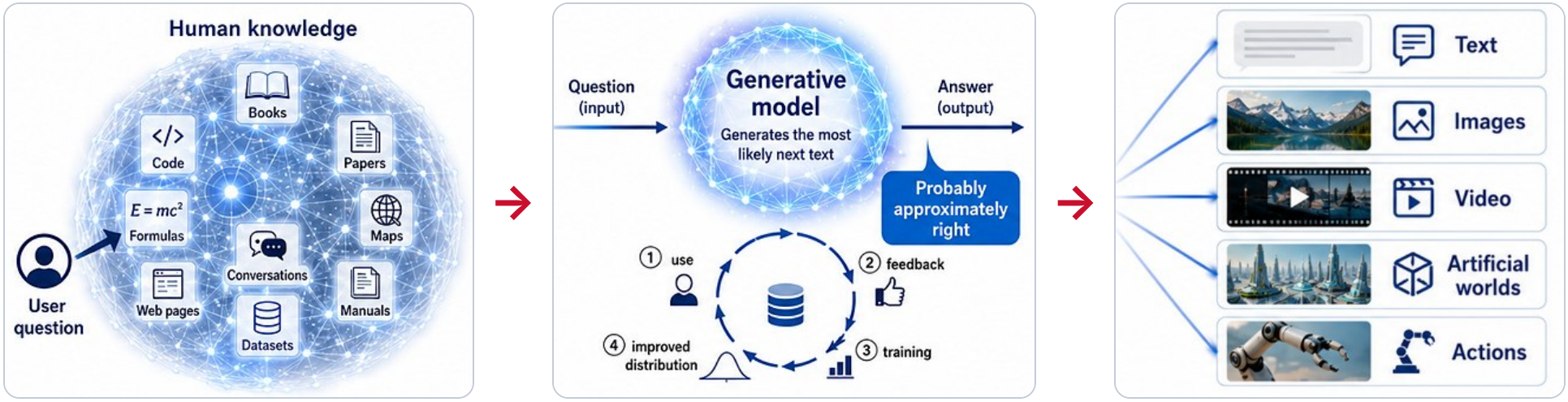
Autonomous driving
Illustrative driving clip

TEST THE RANGE

Name one change to any clip's task or setting that would test how far its competence transfers.

ACT 1 · WHAT A GENERATIVE MODEL RETURNS

A generative model returns a likely continuation without guaranteeing it for the given case



SCOPE OF THE GUARANTEE

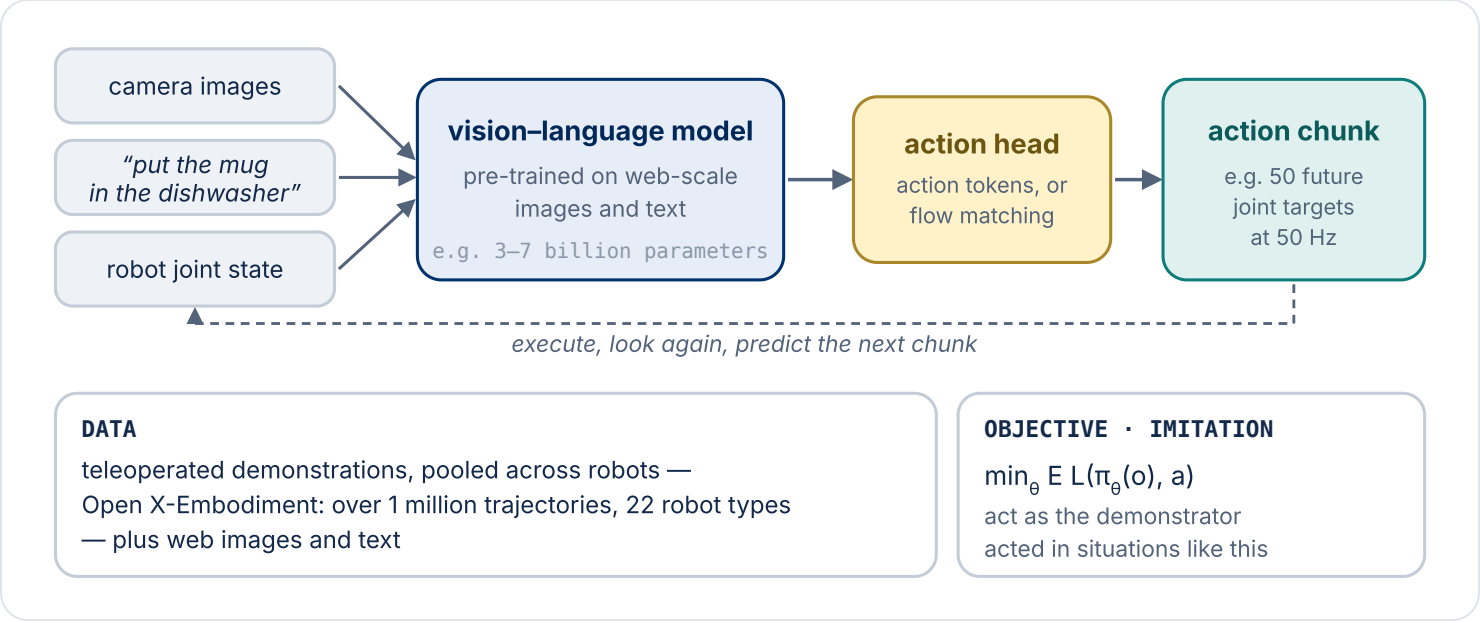
Often right on questions similar to its training data, as measured on benchmarks rather than proven. Nothing certifies the answer to **this** question.

ASYMMETRY BETWEEN TEXT AND ACTION

A wrong paragraph can be read, judged and discarded. A wrong motion has already taken effect; the glass is broken before anyone can judge it.

ACT 1 · HOW AN AI-POWERED ROBOT WORKS

A vision–language–action model, trained on demonstrations, maps images and instructions to action chunks



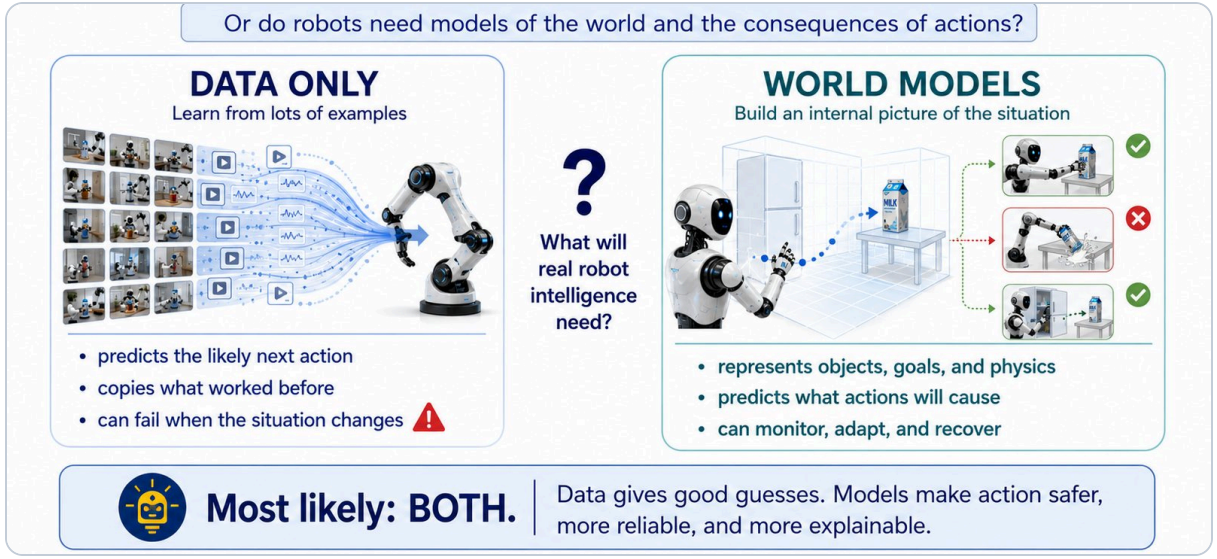
THE TELEOPERATOR AS DATA SOURCE

The person teleoperating the robot supplies the information processing once, as demonstrations; the policy compresses it and replays it in new situations.

Diagram drawn for this lecture — after RT-2 (2023), OpenVLA (2024), π_0 (2024), $\pi_{0.5}$ (2025), Open X-Embodiment (2023)

ACT 1 · IS DATA ALL YOU NEED?

Learned actions and checkable predictions can work together



RUNG	QUESTION	WHAT ANSWERS IT
Seeing	Which action usually follows this observation?	imitation: p(action | observation)
Doing	What will happen if I do this?	a world model of the consequences
Imagining	Why did it work? Would it still work if Eva stood?	a mechanism hypothesis + evidence from the run

TWO QUALIFICATIONS

A world model can itself be **learned**, and a policy may contain one implicitly. The relevant criterion is whether its predictions are **checkable and scoped**, not whether the model was written or learned.

AI-generated illustration · ladder after Pearl & Mackenzie, *The Book of Why* (2018)

ACT 2 · THE QUESTION FOR THIS COURSE

A fuller mug tests whether Eva's coffee plan transfers

Illustrative changed case: Eva accepts the white mug; same robot, route and turn, but more coffee.

Illustrative run · shown success

Half-full white mug

Served without spilling.

→

ILLUSTRATIVE CHANGED CASE · PREDICTION

Nearly full white mug

Spill possible if the turn is unchanged.

The first run supports that case. It does not establish success across fill levels.

Question adapted from James DiCarlo, *What would it mean to understand intelligence?* (CBMM, 2022)

ACT 2 · THE FOUNDATION

Representations expose a model's constraints so algorithms can solve the task

For Eva’s accepted white mug, the robot must represent its fill, the route and Eva’s current position. Winston gives that work a computational chain.

1 · MODEL
Eva’s coffee task
Liquid motion, grasp and handover.

2 · REPRESENT
Current situation
Fill level, mug pose and Eva’s location.

3 · EXPOSE CONSTRAINTS
Carry without spilling
Secure grip, reachable route and limited tilt.

4 · COMPUTE AND REGULATE
Serving method
Choose a route, monitor the mug and adjust motion.

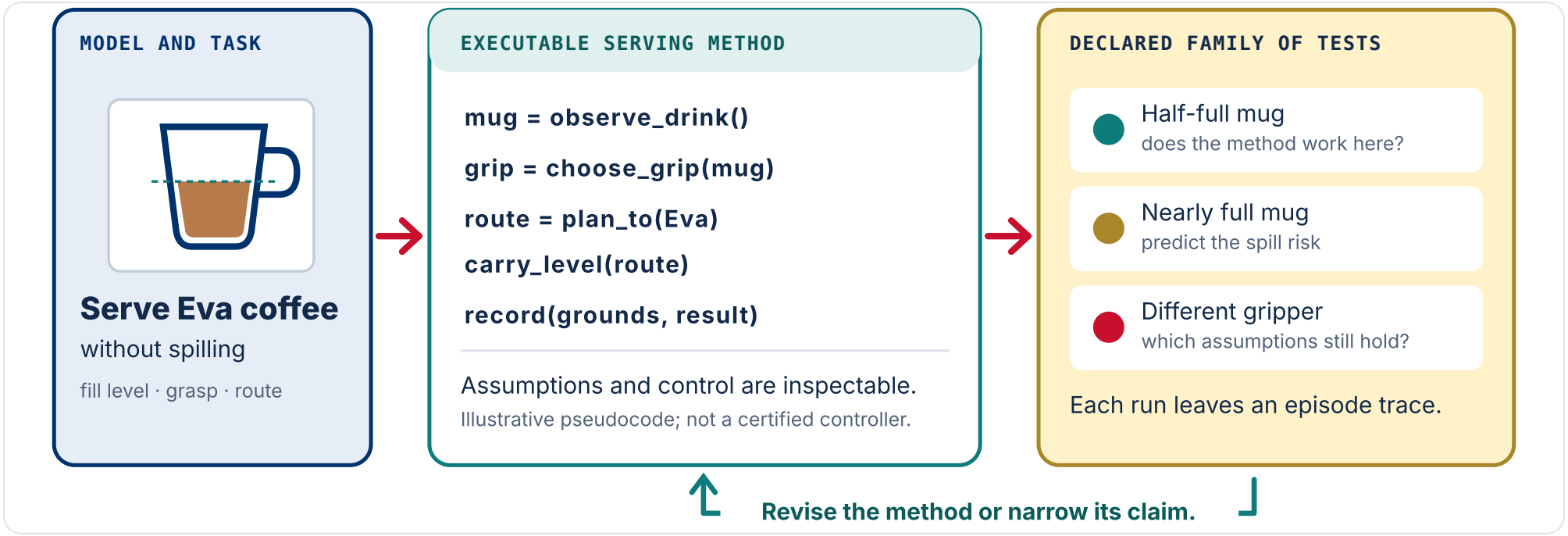
World → model: compare the observed liquid motion and delivery with the prediction. A mismatch calls for a revised model, representation or control choice.

Nilsson, *Reconsiderations* (2005), p. 37 · Winston, MIT 6.034 Lecture 1 (2010)

ACT 2 · HOW WE TEST UNDERSTANDING

Building a working program tests whether our account explains how the task is solved

Eva's goal stays; the mug, fill level and handover can change. A serving program must adapt and be testable.



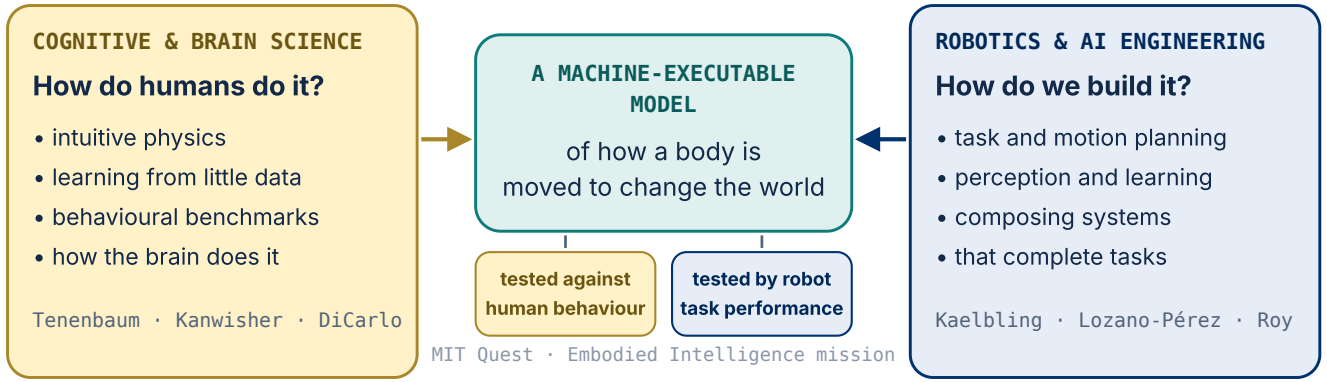
Which assumptions and observations let this program solve a **declared family** of cases?

Illustration drawn for this lecture · Feynman, *Engineering & Science* (2002) · Sussman, *The Legacy of Computer Science* (2004)

ACT 2 · HOW UNDERSTANDING IS PURSUED

Cognitive science and robotics work toward a model of intelligent agency

"How does human intelligence work, in biological as well as in engineering terms?" — MIT Intelligence Quest, 2018



WHAT THIS COURSE MAKES EXPLICIT

Models can be tested against **human behaviour** and **robot task performance**. We also ask whether a robot's commitments are grounded for **this** case and leave evidence others can check.

Diagram after MIT Intelligence Quest and DiCarlo (2022) · Craik (1943) · Battaglia, Hamrick & Tenenbaum (2013); course focus: AICOR

ACT 2 · THE GUIDING QUESTION

Understanding a robot site means answering five kinds of question

Eva's changed coffee episode is the worked case. Apply the same five questions to production and laboratory tasks.

T1
Explain
Why no spill: a slow turn or a half-full mug?

T2
Predict
Will a nearly full mug spill on this route?

T3
Design
Would another gripper or a tray change the carry?

T4
Bound
For which mugs, fills and routes does it work?

T5
Check
Did the record capture Eva's answer, motion and outcome?

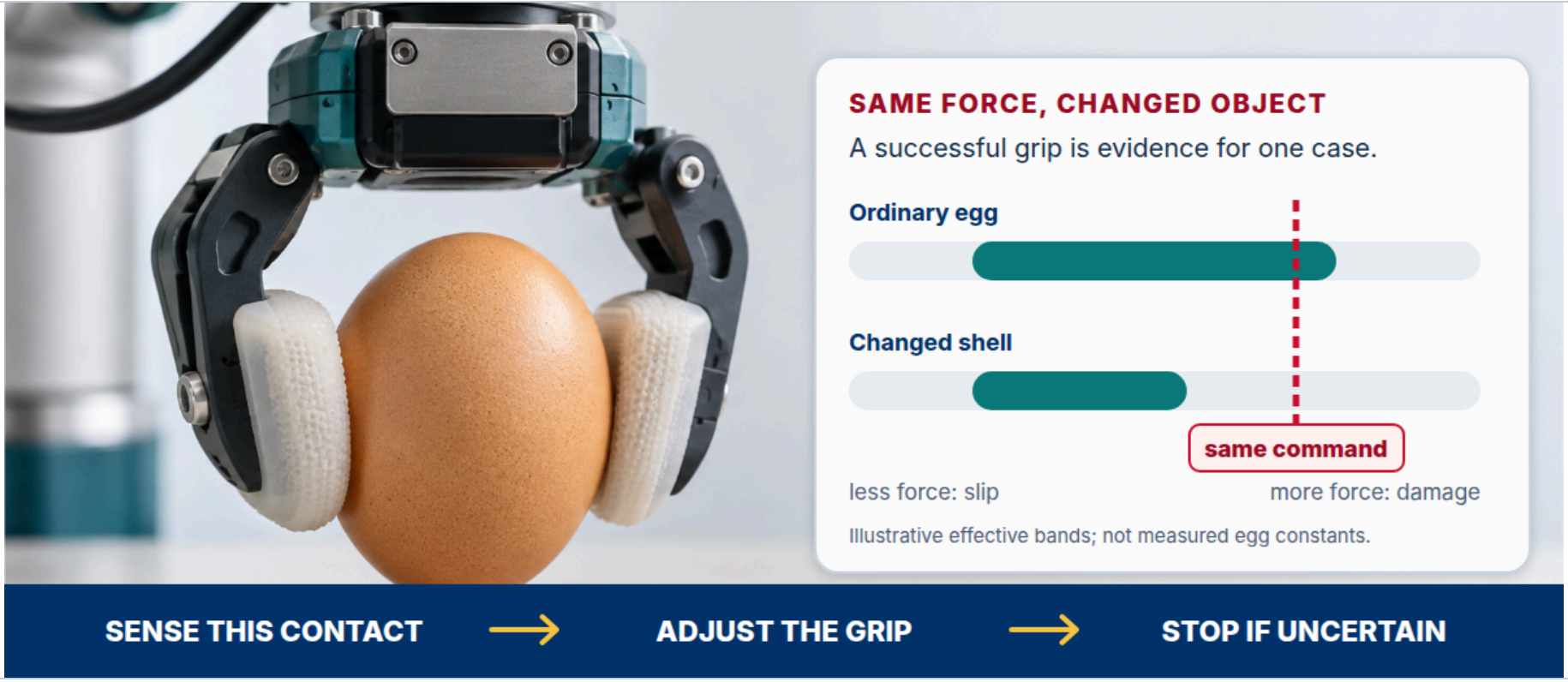
TRANSFER TO NEW CASES

The tests form one cycle: explain a run, predict a change, improve the design, state its scope and let others check the evidence.

PAIRS · 2 MIN Task: choose one site. For each of T1–T5, formulate one question that a video of the site working could NOT answer.

ACT 3 · ONE SMALL PHYSICAL DECISION

A grip that worked before may fail on this egg



THE BASIC IDEA

A good action uses evidence about **this object** and checks contact as it happens.

AI-generated scene with schematic annotations · force regions illustrative, not measured

PAIR TASK · FIVE MINUTES

What must the robot know and do to bring Eva her coffee safely?

Her blue mug is unavailable; she may accept white. The fill level and her position can change.

2 min Individually: list everything the robot must **know** and **do** (at least eight items).

2 min In pairs: merge the lists and group the items that belong together.

1 min Label each group with one or two words.

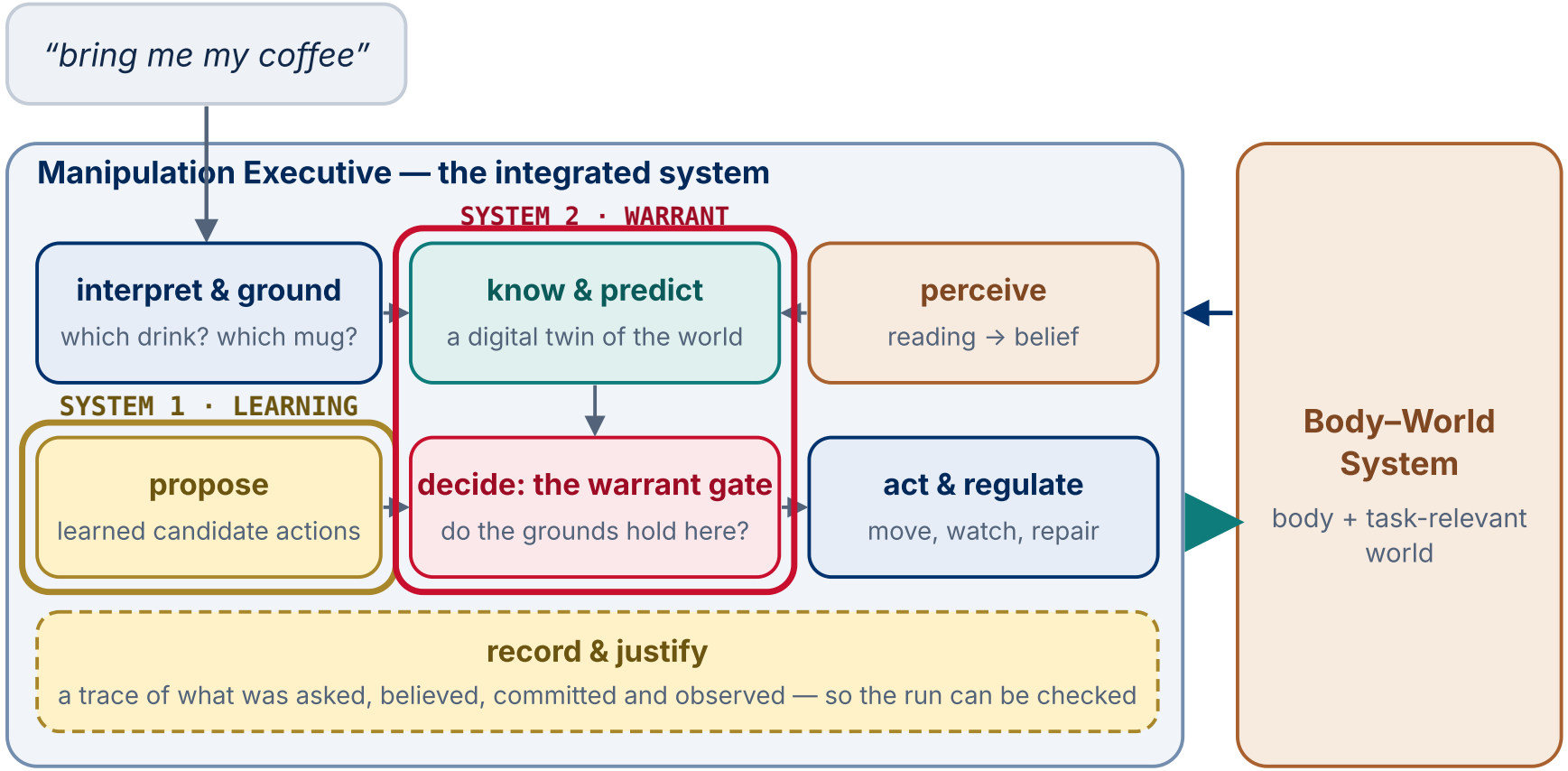
USE OF THE LISTS

The next part builds the system from these lists; the groups become the parts of an integrated intelligent system.

ACT 4 · WHAT “INTEGRATED” MEANS

An integrated intelligent system supplies, as one system, what the human operator supplied

Eva’s goal stays; the accepted mug, safe carry and handover must be revised as the situation changes.



ACT 4 · THE WARRANT GATE, IN EVA'S KITCHEN

A commitment check compares evidence with a declared criterion

GROUNDED CRITERION
Eva has accepted the proposed mug.

AVAILABLE EVIDENCE
Blue mug unavailable; white-mug acceptance unknown.

ROUTE
Ask Eva; then reassess fill, route and handover.

THE PROPOSAL
"Take a mug from the cupboard, fill it with hot coffee, and carry it to Eva's table."
From a learned policy, a planner or a foundation model; the gate is indifferent to the source. It decides whether the proposal is licensed to move the body.

× **grounded**
has Eva accepted the proposed mug? (white-mug acceptance is unknown)

✓ **feasible**
can this body reach the cupboard and carry a full mug level?

✓ **safe**
is hot coffee near Eva within bounds (fill level, speed, distance)?

✓ **monitorable**
can it watch the coffee's level and Eva while it carries?

✓ **repairable**
if Eva stands up or it spills, is there a way to recover?

✓ **recordable**
will the run leave a record that can be checked afterwards?

1 OF 6 CHECKS FAIL — THE FIRST FAILING CHECK ROUTES THE PROPOSAL

Ask: "Your blue mug is in the dishwasher. Is the white one all right?"

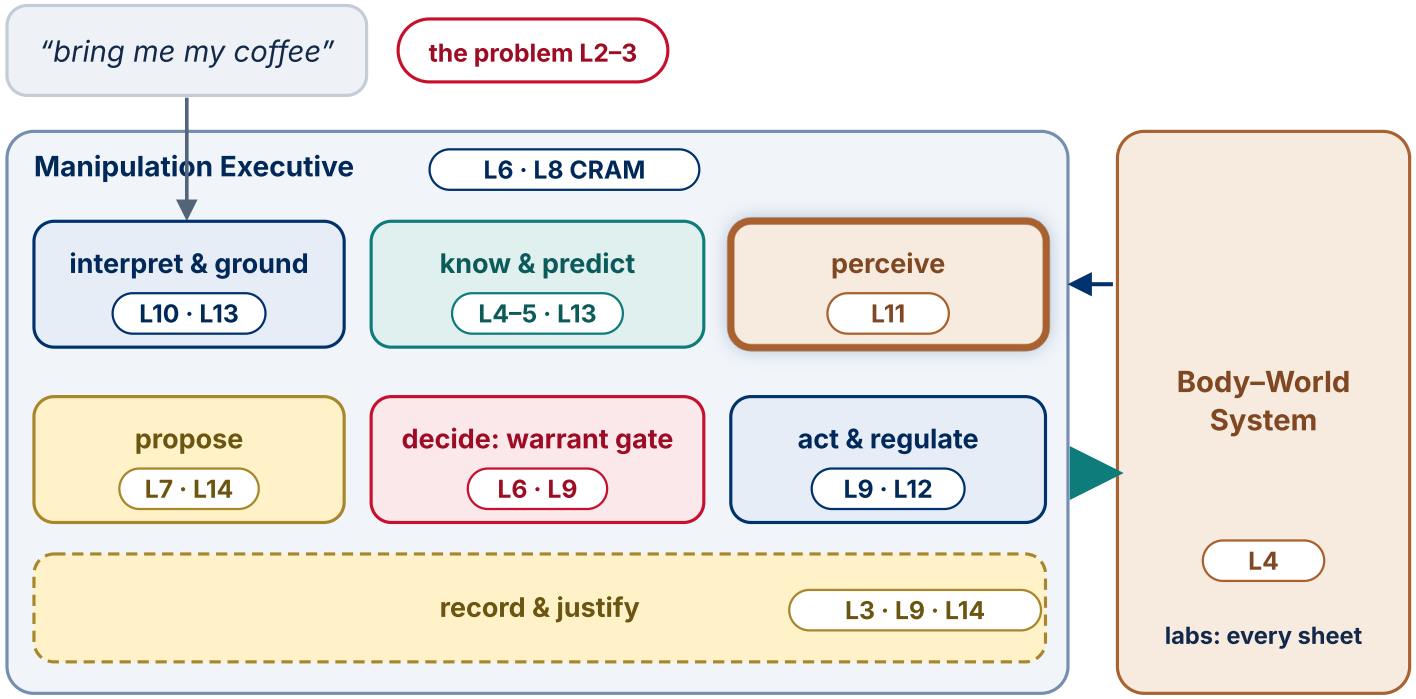
Grounding fails: the blue mug is unavailable, and Eva has not accepted the white one. Ask before committing.

WHAT A PASSED CHECK MEANS
A passed check means available evidence meets this action's declared criterion; uncertainty remains. If Eva stands, pause and check the handover again.

Interactive model — adapted from the AICOR book companion, Ch 7: the warrant gateway

ACT 5 · THE COURSE

The integration diagram is the map of the semester



LECTURE 11

Perceive

`reading → belief`

Turning sensor readings into beliefs about this scene, and keeping the twin consistent with the world. The lecture treats perception as a task-directed process the executive asks questions of, not as a stand-alone recognizer: what must be seen, how sure, and for which action?

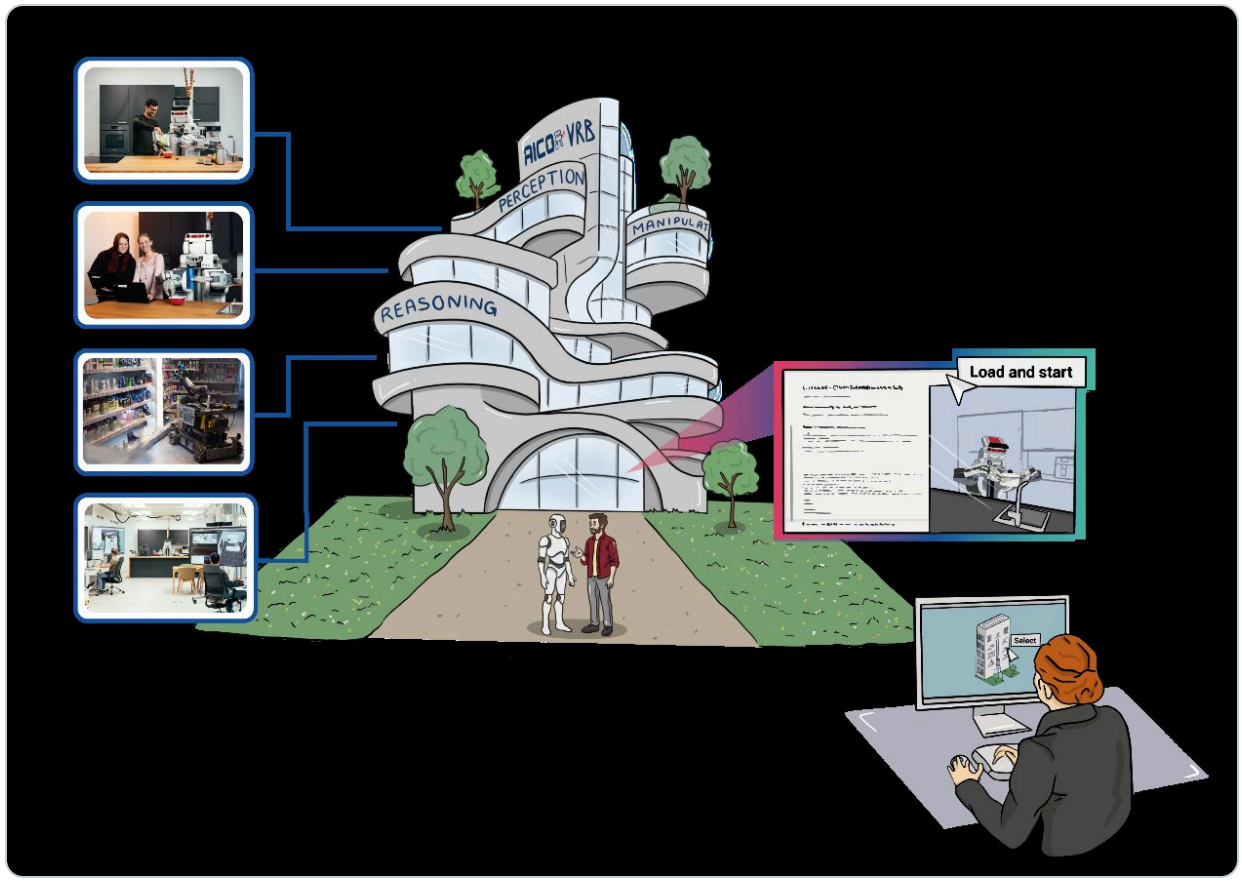
Lecture 11: perception with RoboKudo · **Lecture 5**: the twin as belief state

Click another box.

Each box shows which lecture builds it and which question it addresses.

ACT 5 · THE LABORATORY ENVIRONMENT

Our course labs start with a working Python setup



VIRTUAL RESEARCH BUILDING

Our planned lab environment lets students vary a robot task and compare predicted with observed outcomes. Access details for the first task will come through the course page.

FIVE-MINUTE SETUP CHECK · RUN IT NOW

From the Lecture 1 folder: `python3 tools/check_env.py` (Windows: `py tools/check_env.py`). Note your support code. If it fails, pair with a neighbor after three minutes and give the code to the TA.

ACT 5 · FROM OUR LAB · ONE PART AT A TIME

interpret | **know & predict** | perceive | propose | decide | act | justify

A digital twin lets the robot simulate the consequences of an action before executing it

PREDICT What could a robot ask this model before it goes to fetch Eva's mug?



A digital twin of a real apartment lab — rooms, furniture, appliances, and a robot acting in it
IAI · Bremen

SEMANTIC DIGITAL TWIN

One model joining **what things are** (a dishwasher is a container, its door opens) with **how they move** when touched.

THREE QUESTIONS IT ANSWERS

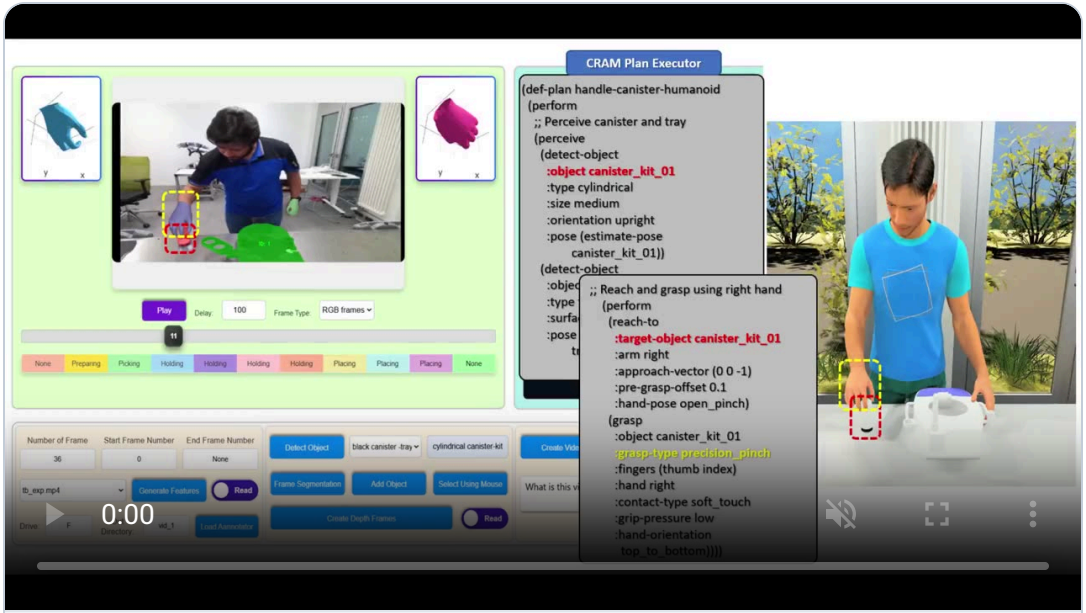
query: where is her mug? (in the dishwasher)
simulate: what happens if the robot pulls here?
predict: what should the robot observe if its belief is correct?

ACT 5 · FROM OUR LAB · ONE PART AT A TIME

interpret | know & predict | perceive | **propose** | decide | act | justify

Robots can extract executable plans from videos of human activity

PREDICT The video shows a person handling a canister. What must the robot extract besides the hand's path?



Video → action designator — a demonstration segmented into actions and written out as an executable plan

IAI · Bremen

ACT 5 · FROM OUR LAB · ONE PART AT A TIME

interpret | know & predict | perceive | propose | **decide** | **act** | justify

One generalized plan runs across robots and kitchens

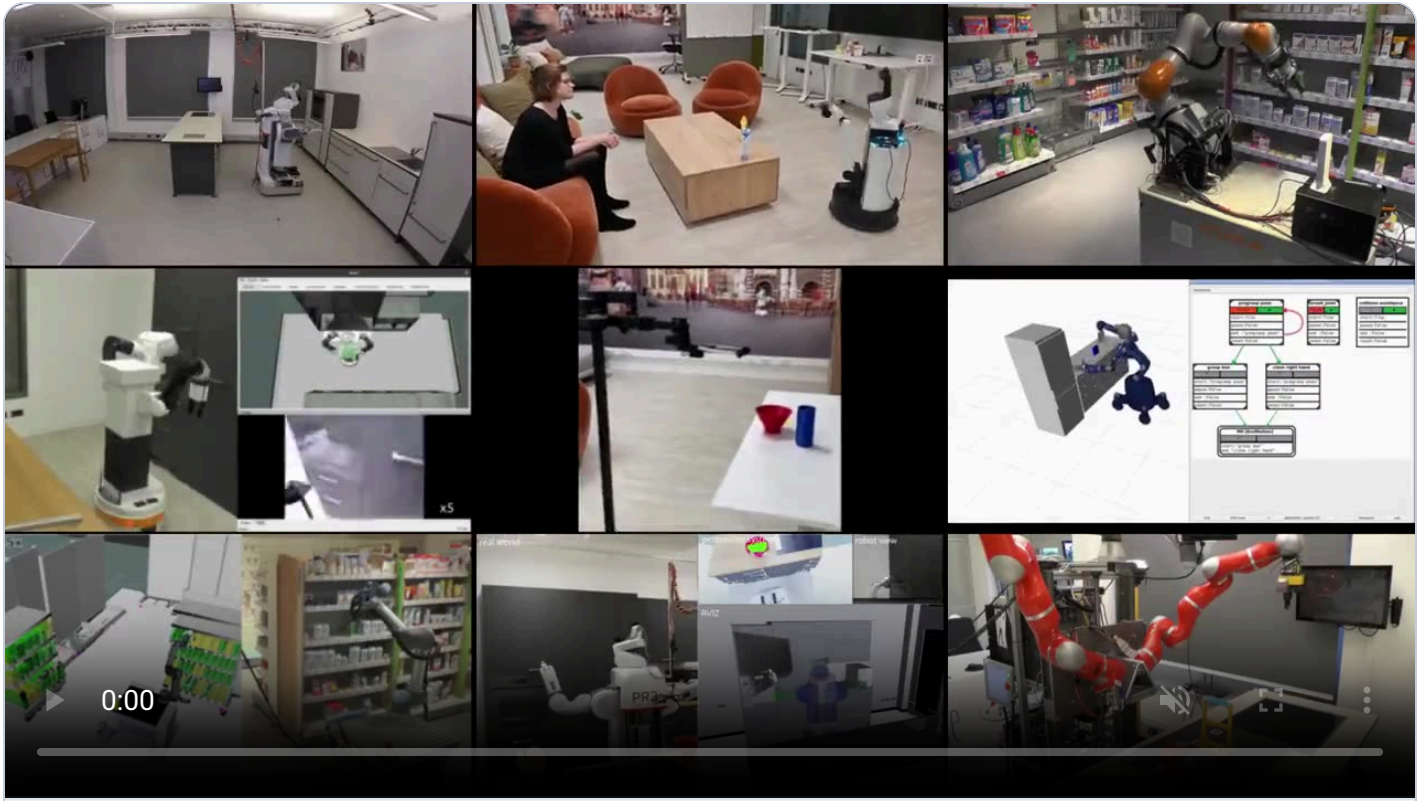
def-plan fetch&place (*?object*, *?destination*)

1. **with-robot-at-location** (a location
   (visible *?object*)
   (reachable *?object*))
   perform (an action
   (type fetching)
   (object *?object*))

2. **with-robot-at-location** (a location
   (visible *?destination*)
   (reachable *?destination*))
   perform (an action
   (type placing)
   (object *?object*)
   (destination *?destination*))

UNDERLYING PRINCIPLE

The plan states **what** must hold (the mug visible and reachable) and leaves **where** and **how** to be inferred for each robot and kitchen.

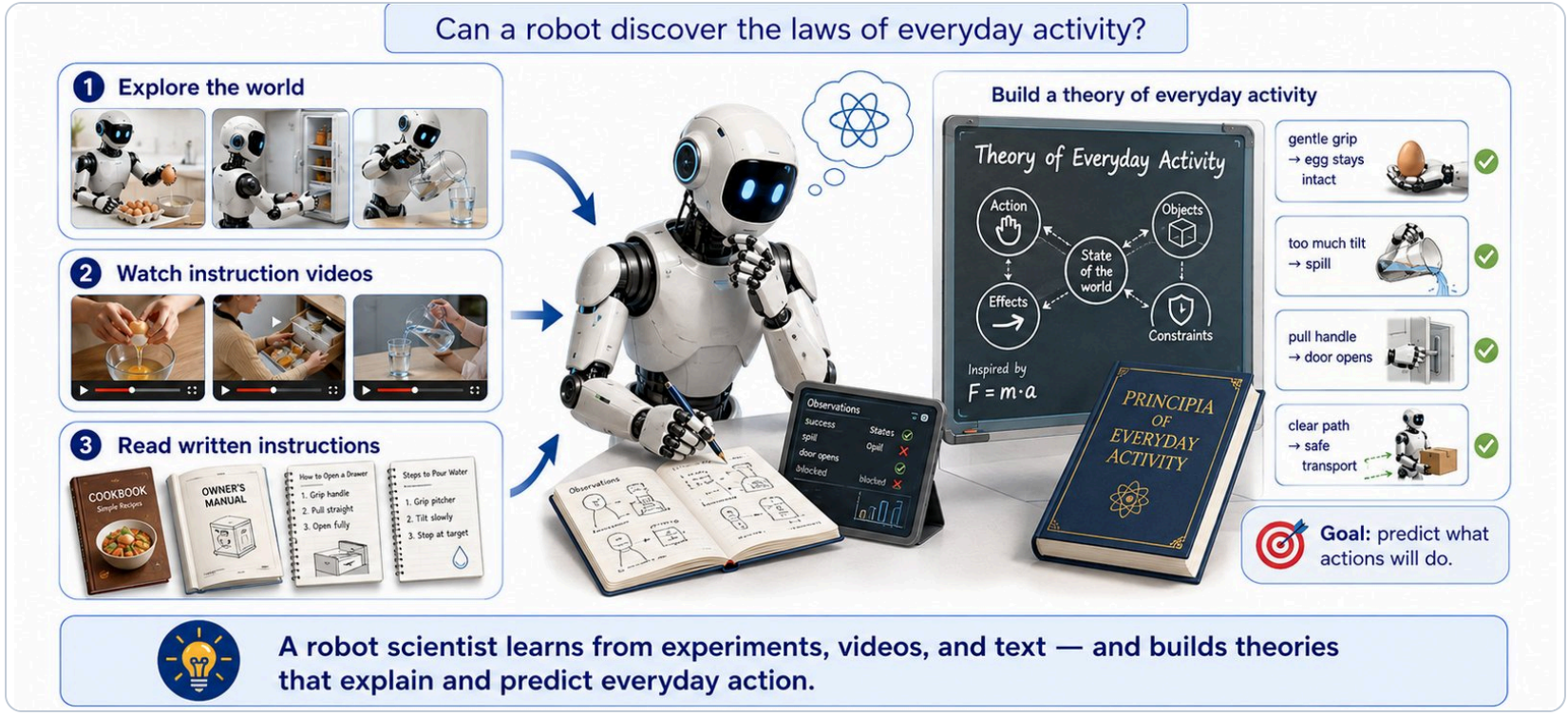


The same plan on different robots, in apartments, a kitchen, a retail store and in simulation

IAI · Bremen · CRAM

ACT 5 · WHERE WE WANT TO GO

The long-term goal is a robot scientist that learns and records how everyday activity works



Beyond performing the task, such a robot learns from experiments, videos and text why actions succeed, and records this so that it can be checked.

AI-generated illustration

CLOSE · COMPARISON WITH THE WARM-UP ANSWERS

A success video supports a claim within demonstrated conditions

Q1 Factory repetition, learned manipulation, autonomous help in an unfamiliar home: which today?

→ Robots repeat factory tasks and learned policies handle diverse objects, including in unseen homes. Assisting Eva autonomously with grounds for each action and a checkable record remains a research goal.

Q2 What does a generative model return?

→ A proposal: a plausible continuation sampled from what it learned, often right on cases like its training data but certified for none.

Q3 100 successes on video: can the robot do the task?

→ **Shown conditions:** one hundred successes support success there. **Untested range:** a wider competence claim needs a declared range, varied tests, checked outcomes and failure evidence.

EXIT TICKET · INDIVIDUAL · STOP AFTER FOUR MINUTES

A changed case tests the scope of a robot claim

Eva still wants coffee. One condition changes. Write one paragraph.

1 · CHANGE

Change one condition.

2 · ADAPT

Revise the action to reach her goal; predict one possible failure.

3 · CHECK

State what evidence would justify that revised action.

4:00 · stop and collect

Before next time

Read AICOR book: Preface, *The AICOR Framework at a Glance*, and the opening of Chapter 1

Do Exercise Sheet 1: first VRB lab; mapping the pair-task list onto the integration diagram

NEXT · LECTURE 2

Analysis of the body motion problem: from a verbal request to body motion

AICOR book, Chapter 1

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AICOR · Institute for Artificial Intelligence · University of Bremen